



KTU NOTES

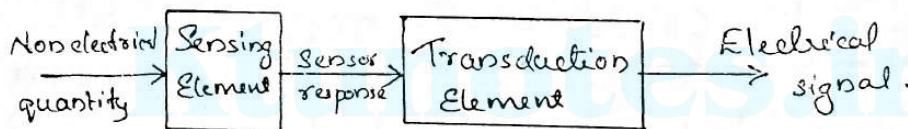
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TRANSDUCERS

- It is a device which converts a physical quantity into proportional electrical signal.
- The electric signal produced may be a voltage, current or frequency
- The process of transforming signal from one form to other is called transduction
- Electrical transducers consist of two parts,
 - sensing or detecting element and
 - transduction element.



↳ Sensor or detector is a device that produces a measurable response to a change in a physical condition

↳ The transduction element transforms the output of the sensor to a electrical quantity.

↳ Sensing element is that part of transducer which responds to a change in a physical phenomenon and its output is then fed to transduction element.

- Thus a transducer in general form may be defined as a device which converts energy in one form to other.

Classification of Transducers

A) Electrical

The transducers can be classified as,

i) on the basis of transduction form used:-

The transducers are classified on the basis of transduction principle as resistive, inductive, capacitive etc. depending on how they convert the input quantity into resistance, inductance or capacitance respectively.

They can be classified as eddy current, thermoelectric, magnetostrictive, and optical.

ii) Primary and Secondary transducers:-

Consider an example of measurement of pressure using bourdon tube and LVDT

The bourdon tube acting as a primary detector senses the pressure and converts the pressure into displacement of its free end. The displacement of free end moves core of a LVDT (linear variable differential transformer) which produces an output voltage proportional to displacement of the free end which in turn is proportional to the pressure.

Thus there are two stages of transduction, first the pressure is converted into displacement by bourdon tube, called primary transducer and the displacement is converted to voltage by LVDT called as secondary Transducer.

11) Passive and Active Transducers:-

(2)

Passive transducer derive the power required for transduction from an auxiliary power source. They also derive part of power required for conversion from the physical quantity under measurement. They are also known as externally powered transducers.

eg:- Resistive, inductive, capacitive

Active transducers are those which do not require an auxiliary power source to produce their output. They are also known as self generating type. as the energy required for production of output signal is obtained from physical quantity being measured.

12) Analog and digital transducers:-

Analog transducer convert the input quantity into an analog output which is a continuous function of time. eg. strain gauge, LVDT

Digital transducer convert the input quantity into an electrical output which is in the form of pulses.

13) Transducers and inverse transducers:-

Transducers convert a non-electrical quantity into an electrical quantity.

An Inverse transducer is a device which converts an electrical quantity into a non-electrical quantity. eg:- piezoelectric crystal as it converts voltage applied to it to displacement.

LVDT (Linear variable Differential transformer)

- It is an inductive transducer to transducer to translate the linear motion into electrical signal

- used for measurement of linear displacements

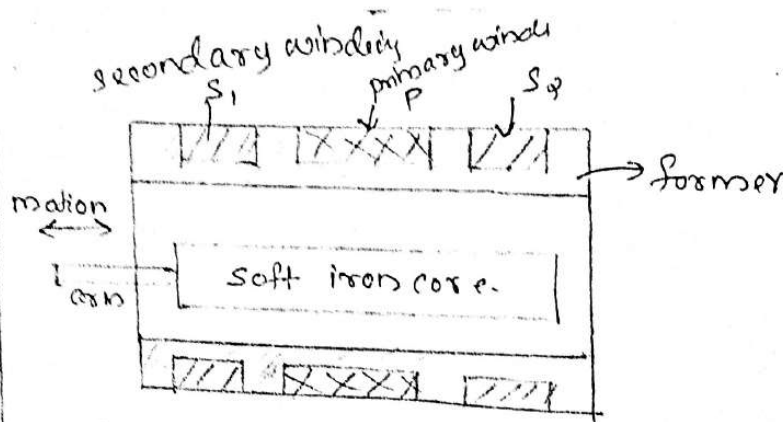
construction:-

It consist of a single primary winding P and two secondary windings S_1 and S_2 wound on a cylindrical former.

The secondary windings have equal number of turns and are identically placed on either side of the primary winding. The primary winding is connected to an alternating current source.

A movable soft iron core is placed inside the former.

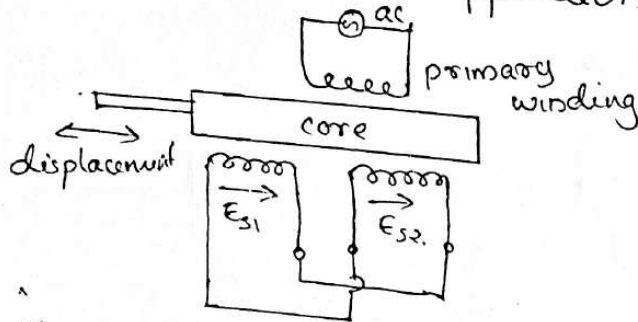
A movable soft iron core is placed inside the former. The displacement to be measured is applied to the arm attached to soft iron core



Working:-

- Since the primary winding is excited by an alternating current source, it produces an alternating magnetic field which in turn induces alternating current voltages in the two secondary windings.

- The output voltage of the secondary S_1 is E_{s1} and that of secondary S_2 is E_{s2} . In order to convert the outputs from S_1 and S_2 into a single voltage signal, the two secondaries S_1 and S_2 are connected in series opposition.



Differential o/p $E_o = E_{s1} - E_{s2}$

When core is at null position, flux linking with both the secondary winding is equal and hence equal emfs are induced in them. Thus at null position, $E_{s1} - E_{s2} = 0$

(9)
If core is moved to the left of the NULL position, more flux links with the winding S_1 and less ^{with} winding with S_2 . Therefore output voltage E_{s1} of secondary winding S_1 is more than E_{s2} output voltage of secondary winding S_2 .

$\therefore$ o/p voltage $E_o = E_{s1} - E_{s2}$ is in phase with primary voltage.

If the core is moved to right of null position the flux linking with winding S_2 becomes larger than that linking with winding S_1 . Therefore E_{s2} will be larger than E_{s1} .

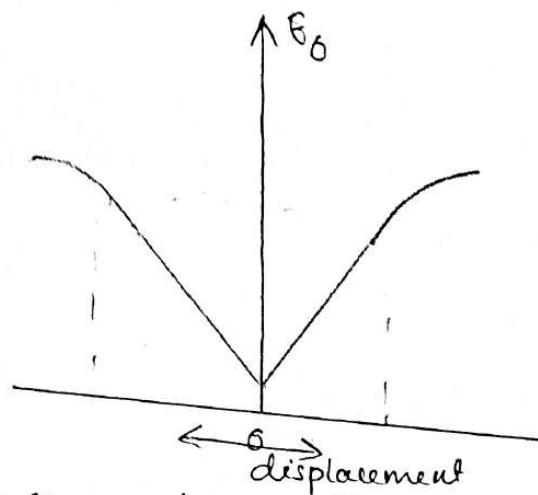
$\therefore$ output voltage $E_o = E_{s1} - E_{s2}$ is 180° out of phase with primary voltage.

The amount of voltage change in either secondary winding is proportional to the amount of movement of core. Hence we have an indication of amount of linear motion.

By comparing magnitude and phase of the output voltage with that of source, amount and direction of movement of core and hence displacement can be measured.

Output characteristics:-

Figure shows the variation of output voltage versus displacement for various position of core.



The curve is practically linear for a limited range of displacement from the null position. Beyond this range, curve starts to deviate from straight line.

Although output voltage V_o at null position is should be zero, it is observed that small amount of voltage appears across output terminals due to

- i) incomplete electrical and magnetic balance between output voltage at Null position
- ii) due to stray magnetic field or temp

Advantages

- 1) high sensitivity
- 2) Infinite resolution
- 3) less friction
- 4) low power consumption
- 5) good linearity

Disadvantages

- 1) high displacement required to generate high voltages.

rates approach zero, the slippage may amount to 100%, i.e. the turbine stalls and registers zero output. Therefore, at low flow rates, there is a possibility of large errors.

29.40.2. Electromagnetic Flow meters

Electromagnetic flow meters are particularly suitable for the flow measurements of slurries, sludge and any electrically conducting liquid.

A schematic diagram of an electromagnetic flowmeter is shown in Fig. 29.84. It consists basically of a pair of insulated electrodes buried flush in the opposite sides of a non-conducting, non-magnetic pipe carrying the liquid whose flow is to be measured. The pipe is surrounded by an electromagnet which produces a magnetic field across a magnetic field. Therefore, voltage is induced across the electrodes. This voltage is given by : $E = B l v$ where B = flux $\times$ density ; Wb/m^2 , l = length of conductor = diameter of pipe ; m, and v = velocity of conductor (flow) ; m/s

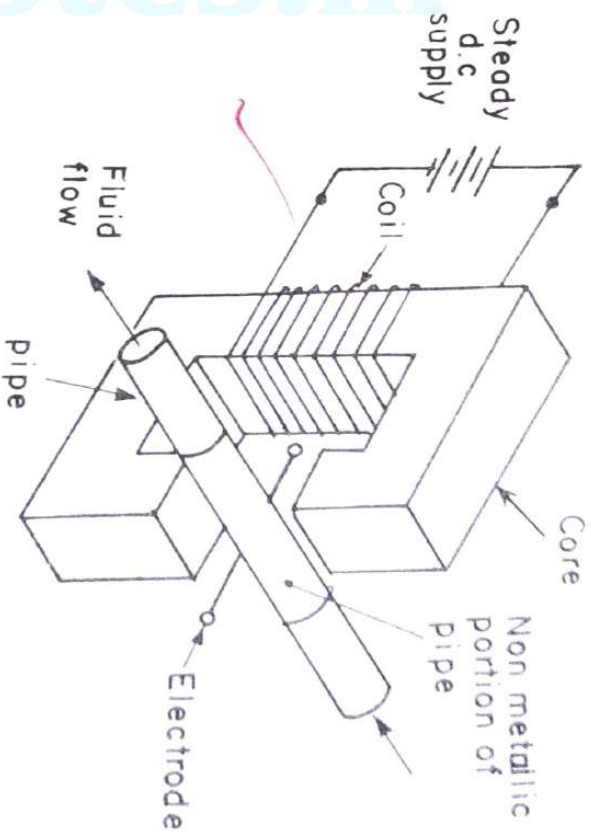


Fig. 29.84. Electromagnetic flow meter.

Thus, assuming a constant magnetic field, the magnitude of the voltage appearing across the electrodes will be directly proportional to velocity.

Non-conducting pipe has to be used as the output voltage gets short circuited if metallic pipes are used. This is true when liquids of low conductivity are being measured. But when liquids of high conductivity are measured the short circuiting has no effect. Stainless steel pipes can then be used. The voltages produced are small specially at low flow rates. Therefore, the meter relies greatly on a high gain amplifier to convert the induced voltage into a usable form.

Advantages : (i) The electromagnetic flowmeters may be manufactured to measure flow in pipes of any size provided powerful magnetic field can be produced.

(ii) The major advantage from a fluid handling point of view is that with the electro-magnetic flow meter there is no obstruction to flow that may cause pressure drops.

(iii) The output (voltage) is linearly related to the input (flow rate).

(iv) The output is unaffected by changes in characteristics of liquid such as viscosity, pressure, and temperature.

Limitations : (i) The operating costs are high particularly if heavy slurries are handled.

(ii) The conductivity of the liquid being metered should not be less than $10 \mu \Omega/\text{m}$. As a general rule it will be found that most aqueous solutions are adequately conductive while majority of hydrocarbon solutions are not sufficiently conductive.

Comparison of D.C. and A.C. Excitation. The magnetic field used in electromagnetic flow meters can either be d.c. or a.c. giving rise to d.c. or an a.c. output signal respectively. The d.c. is used in limited applications due to many reasons :

Advantages :

1. No obstruction to flow.
2. Can handle slurries, greasy materials and liquids containing suspended matter.
3. High accuracy and reliability.
4. The output is unaffected by changes in characteristics of liquid such as viscosity, pressure and temperature.
5. The output (voltage) is linearly related to the input (flow rate).
6. Bidirectional flow can be measured by reversing connections which can be done manually or automatically.
7. The mass flow rate can be found by measuring the density and then multiplying the output signals for indication of mass flow rate.
8. The voltage output of these meters is proportional to average velocity and hence it does not matter whether the flow is laminar or turbulent.

Disadvantages :

1. The operating costs are high particularly if heavy slurries are handled.
2. Its use is limited to fluids having conductivity at least of the order of 0.05 $\mu\text{mho/cm}$.
3. They must be full at all the times because it sees velocity as analogues to volume flow rate.

Application. This type of meter in its various forms has been widely used for metering corrosive acids, cement slurries, sewage, paper pulp, detergents etc. which present difficult handling problems.

12.3.9. Ultrasonic Flow Meters

The working principle of these meters is based on an apparent change in velocity of propagation of sound waves in a fluid with a change in the velocity of the fluid flow. The method uses transmitters and receivers of acoustic energy. "Piezoelectric crystals" are used as both transmitters and receivers.

Fig. 12.42 illustrates an ultrasonic flow meter for measurement of flow.

Refer to Fig. 12.42(i) : Travel time difference method-Single transmitter-receiver system.

-When flow velocity (v) is zero, the transit time of the pulse from the transmitter to the receiver is given by :

$$t_0 = \frac{d}{c} \quad \dots(12.42)$$

where,

d = Distance between the transmitter and receiver, and

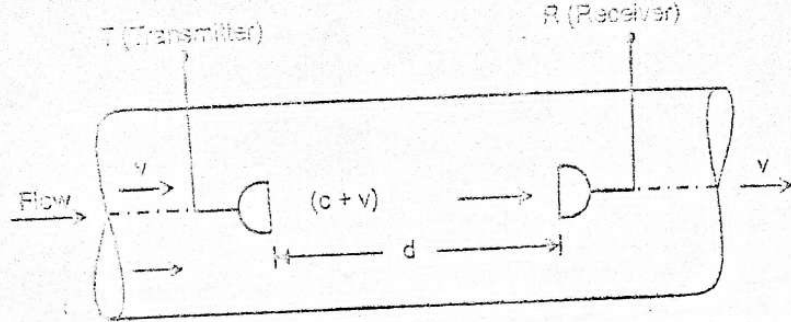
c = Velocity of sound in fluid.

Now, when the fluid moves with velocity v , the transit time becomes :

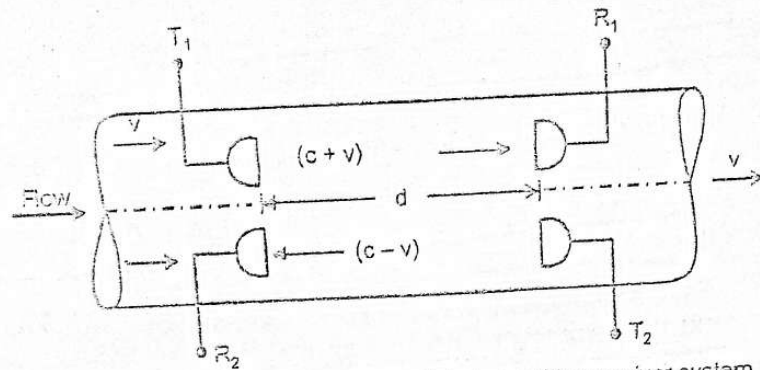
$$t = \frac{d}{c+v} \quad \dots(12.43)$$

$$= \frac{d(c-v)}{(c+v)(c-v)} = \frac{d(c-v)}{c^2 - v^2}$$

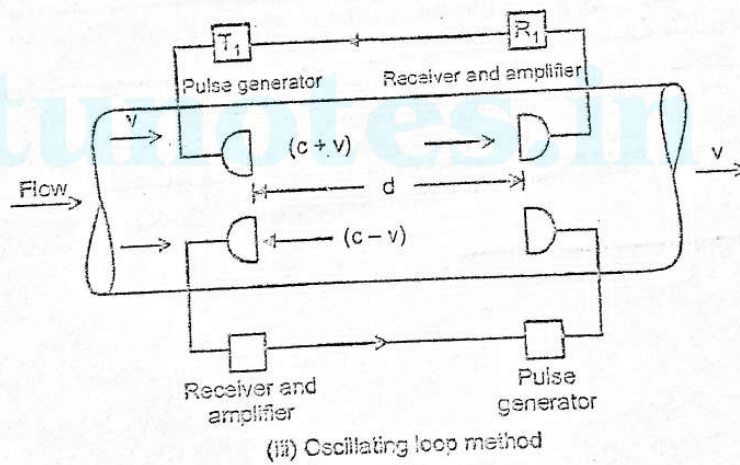
...Multiplying numerator and denominator by $(c-v)$



(i) Travel time difference method—single transmitter-receiver system



(ii) Travel time difference method—Twin transmitter-receiver system



(iii) Oscillating loop method

Fig. 12.42. Ultrasonic flow meter.

For any reasonable flow rate in liquids, the relationship $v \ll c$ holds and negligible error in writing, then

$$t = \frac{d(c-v)}{c^2}$$

$$= \frac{d}{c} \left(1 - \frac{v}{c} \right)$$

or,

$$t = t_0 \left(1 - \frac{v}{c} \right)$$

or,

$$t_0 - t = t_0 \frac{v}{c}$$

Let

 $t_0 - t = \Delta t$ (Travel time difference), then

$$\Delta t = t_0 \frac{v}{c}$$

or,

$$\Delta t = \frac{dv}{c^2}$$

$$\left(\because t_0 = \frac{d}{c} \right) \quad \dots(12.45)$$

Refer to Fig. 12.42 (ii) : Travel time difference method-Twin transmitter-receiver method :

Because the measurement of t_0 is generally not provided for in the present measurement, therefore, it is preferable to have additional set of transmitter-receiver system along with the present system [see Fig. 12.42(ii)] to determine the transmit time against the direction of flow. If the t_1 is the transit time along the flow and t_2 the transmit time against the flow, then the travel time difference Δt is given by :

$$\begin{aligned} \Delta t &= t_2 - t_1 \\ &= \frac{d}{c-v} - \frac{d}{c+v} \\ &= \frac{2dv}{c^2 - v^2} \end{aligned}$$

$$(\because v \ll c) \quad \dots(12.46)$$

or,

$$\Delta t = \frac{2dv}{c^2}$$

Thus an output signal proportional to Δt is linear in v for constant c but the calibration constant will be strongly dependent on c (which depends on the temperature and pressure of the flow field).

Refer to Fig. 12.42 (iii) : Oscillating loop method :

In oscillating method, also called frequency difference technique the effect of the sonic velocity is eliminated by arranging each pair of transducer is an oscillating loop [Fig. 12.42(iii)]. A pulse is emitted by the transmitting transducer T_1 and is received by the receiving transducer R_1 after time t_1 . This pulse is amplified and instantaneously fed back to the transmitting transducer for retransmission. This generates a train of pulses in each path whose period equals the acoustic travel time. The reflection frequencies (f_1, f_2) and the frequency difference or beat frequency between two loops ($f = f_1 - f_2$) are given as :

$$f_1 = \frac{1}{t_1} = \frac{c+v}{d}; \quad f_2 = \frac{1}{t_2} = \frac{c-v}{d} \quad \dots(12.47)$$

$$f = f_1 - f_2 = \frac{c+v}{d} - \frac{c-v}{d} = \frac{2v}{d} \quad \dots(12.48)$$

The frequency difference is thus proportional to v and independent of sonic velocity c .

- Since frequency difference is an easily measured quantity, this method has considerable appeal.
- The drawback of this method is that the frequency difference is small, and the long counting interval needed for a given resolution results in slow response time.

Advantages : The following are the advantages and disadvantages of ultrasonic flow meter :

1. Absence of moving parts.
2. No obstruction to flow.
3. Excellent dynamic response.
4. High accuracy (of the order of $\pm 2\%$ of the full scale values).

...(12.49)

25.29. Piezo-electric Transducers. A piezo-electric material is one in which an electric potential appears across certain surfaces of a crystal if the dimensions of the crystal are changed by the application of a mechanical force. This potential is produced by the displacement of charges. The effect is reversible, i.e., conversely, if a varying potential is applied to the proper axis of the crystal, it will change the dimensions of the crystal thereby deforming it. This effect is known as **piezo-electric effect**. Elements exhibiting piezo-electric qualities are called as *electro-resistive elements*.

Common piezo-electric materials include Rochelle salts, ammonium dihydrogen phosphate, lithium sulphate, dipotassium tartarate, potassium dihydrogen phosphate, quartz and ceramics A and B. Except for quartz and ceramics A and B, the rest are man-made crystals grown from aqueous solutions under carefully controlled conditions. The ceramic materials are polycrystalline in nature. They are, basically, made of barium titanate. They do not have piezo-electric properties in their original state but these properties are produced by special polarizing treatment.

The materials that exhibit a significant and useful piezo-electric effect are divided into two categories :
(i) Natural group and (ii) Synthetic group.

Quartz and Rochelle salt belong to *natural group* while materials like lithium sulphate, ethylene diamine tartarate belong to the *synthetic group*.

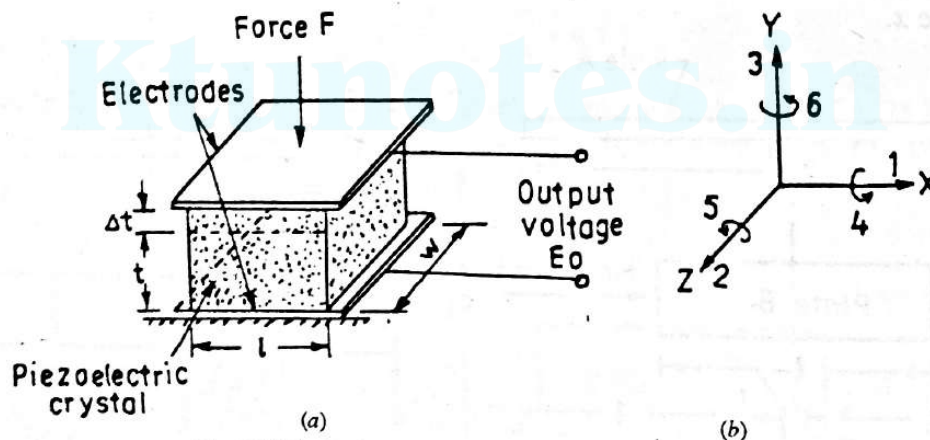


Fig. 25.121. (a) Piezo-electric crystal used for measurement of force.
(b) Axis numbering system for the crystal.

The piezo-electric effect can be made to respond to (or cause) mechanical deformations of the material in many different modes. The modes can be : thickness expansion, transverse expansion, thickness shear and face shear. The mode of motion effected depends on the shape of the body relative to the crystal axis

and location of the electrodes. A piezo-electric element used for converting mechanical motion to electrical signals may be thought as charge generator and a capacitor. Mechanical deformation generates a charge and this charge appears as a voltage across the electrodes. The voltage is $E = Q/C$.

The piezo-electric effect is direction sensitive. A tensile force produces a voltage of one polarity while a compressive force produces a voltage of opposite polarity.

A piezo-electric crystal is shown in Fig. 25.121.

The magnitude and polarity of the induced surface charges are proportional to the magnitude and direction of the applied force F . The polarity of induced charges depends upon the direction of applied force.

Charge $Q = d \times F$ coulomb

$$d = \text{charge sensitivity of the crystal ; C/N : (it is constant for a given crystal)} \quad \dots(25.130)$$

where

$$F = \text{applied force, N,}$$

and

The force F causes a change in thickness of the crystal.

$$F = \frac{AE}{t} \Delta t \text{ newton} \quad \dots(25.131)$$

where A = area of crystal ; m^2 , t = thickness of crystal ; m , and E = Young's modulus, N/m^2 .

Young's modulus $E = \frac{\text{stress}}{\text{strain}} = (F/A) \cdot \frac{1}{\Delta t/t} = \frac{Ft}{A\Delta t} \text{ N/m}^2 \quad \dots(25.132)$

Area $A = wl$ where w = width of crystal ; m , and l = length of crystal ; m .

$\therefore$ From Eqns. 25.130 and 25.131, we have, charge :

$$Q = dAE(\Delta t/t) \quad \dots(25.133)$$

The charge at the electrodes gives rise to an output voltage E_0 .

Voltage $E_0 = Q/C_p \quad \dots(25.134)$

where

C_p = capacitance between electrodes ; F.

Capacitance between electrodes $C_p = \epsilon_r \epsilon_0 A/t \quad \dots(25.135)$

From Eqns. 25.130, 25.134, 25.135,

$$E_0 = \frac{Q}{C_p} = \frac{d F}{\epsilon_r \epsilon_0 A/t} = \frac{d t}{\epsilon_r \epsilon_0} \frac{F}{A} \quad \dots(25.136)$$

But

$$F/A = P = \text{pressure or stress in } N/M^2.$$

$$E_0 = \frac{d}{\epsilon_r \epsilon_0} t P \quad \dots(25.137)$$

$$= g t P \quad \dots(25.138)$$

where

$$g = d/\epsilon_r \epsilon_0 \quad \dots(25.139)$$

' g ' is the voltage sensitivity of the crystal. This is constant for a given crystal cut. Its units are Vm/N .

Now

$$g = \frac{E_0}{t P} = \frac{E_0/t}{P} \quad \dots(25.140)$$

But E_0/t = electric field strength, V/m , Let $\epsilon = E_0/t$ = electric field

$$\therefore g = \frac{\text{electric field}}{\text{stress}} = \frac{\epsilon}{P} \quad \dots(25.141)$$

Now E_0/t is the electric field intensity in the crystal and P is the pressure or the applied stress to the crystal. Therefore, crystal voltage sensitivity, g , can be defined as the ratio of electric field intensity to pressure (or stress). Now $E_0/t = \epsilon$ is the electric field intensity in the crystal and P is the pressure or the stress applied to the crystal. Therefore, crystal voltage sensitivity, g , can be defined as the ratio of the electric field intensity to pressure (or stress). The units of g are Vm/N .

From Eqn. 25.139,

charge sensitivity

$$d = \epsilon_r \epsilon_0 g \quad C/N \quad \dots(25.142)$$

(i) Thickness shear, (ii) Face shear, (iii) Thickness expansion and (iv) Transverse expansion. These modes are shown in Fig. 25.122.

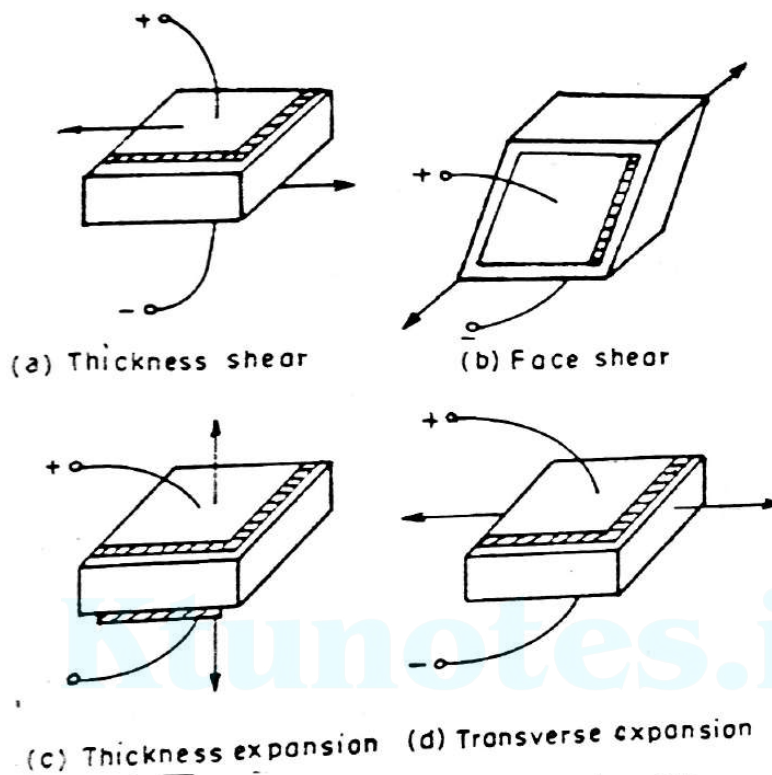


Fig. 25.122. Modes of operation of piezo-electric crystals.

By cementing two crystals together so that their electrical axes are perpendicular, 'benders' or 'twisters' can be produced (See Fig. 25.123). This means that a bending motion applied to a bender produces an output voltage. Similarly a twisting motion applied to a twister produces an output.

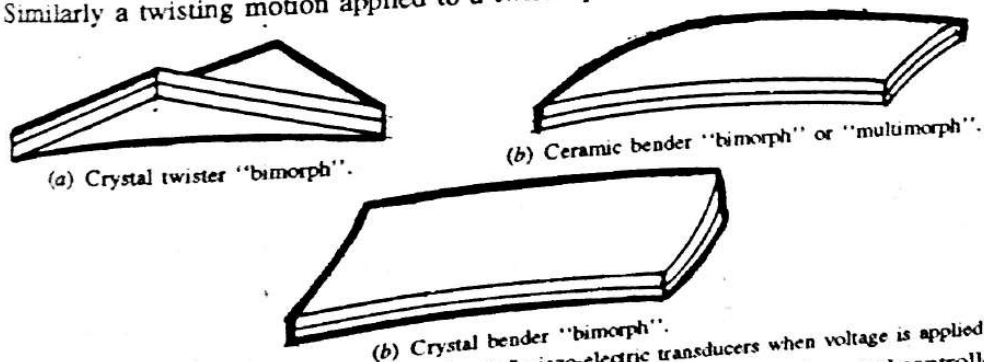


Fig. 25.123. Curvature of "twister" and "bender" piezo-electric transducers when voltage is applied.

Piezo-electric crystals can be used in another mode for force measurement. A crystal controlled electronic oscillator uses a thin plate of quartz. The natural frequency of mechanical oscillation of the plate determines the frequency of electrical oscillation.

of the potentiometer is 100°C and therefore de-rating in power dissipation has to be done on account of increase in temperature. The de-rated power dissipation is 650 mW while the potentiometer is dissipating 667 mW and hence it will fail.

25.16. Strain Gauges. If a metal conductor is stretched or compressed, its resistance changes on account of the fact that both length and diameter of conductor change. Also there is a change in the value of resistivity of the conductor when it is strained and this property is called **piezoresistive effect**. Therefore, resistance strain gauges are also known as **piezoresistive gauges**. The strain gauges are used for measurement of strain and associated stress in experimental stress analysis. Secondly, many other detectors and transducers, notably the load cells, torque meters, diaphragm type pressure gauges, temperature sensors, accelerometers and flow meters, employ strain gauges as secondary transducers.

25.16.1. Theory of Strain Gauges. The change in the value of resistance by straining the gauge may be partly explained by the normal dimensional behaviour of elastic material. If a strip of elastic material is subjected to tension, as shown in Fig. 25.55 or in other words positively strained, its longitudinal dimension will increase while there will be a reduction in the lateral dimension. So when a gauge is subjected to a positive strain, its length increases while its area of cross-section decreases as shown in Fig. 25.55. Since the resistance of a conductor is proportional to its length and inversely proportional to its area of cross-section, the resistance of the gauge increases with positive strain. The change in the value resistance of strained conductor is more than what can be accounted for an increase in resistance due to dimensional changes. The extra change in the value of resistance is attributed to a change in the value of resistivity of a conductor when strained. This property, as described earlier, is known as **piezoresistive effect**.

Let us consider a strain gauge made of circular wire. The wire has the dimensions : length = L , area = A , diameter = D before being strained. The material of the wire has a resistivity ρ .

$\therefore$ Resistance of unstrained gauge $R = \rho L/A$.

Let a tensile stress s be applied to the wire. This produces a positive strain causing the length to increase and area to decrease as shown in Fig. 25.55. Thus when the wire is strained there are changes in its dimensions. Let ΔL = change in length, ΔA = change in area, ΔD = change in diameter and ΔR = change in resistance.

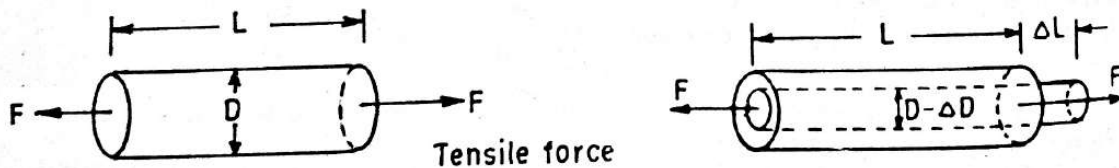


Fig. 25.55. Change in dimensions of a strain gauge element when subjected to a tensile force.

In order to find how ΔR depends upon the material physical quantities, the expression for R is differentiated with respect to stress s . Thus we get :

$$\frac{dR}{ds} = \frac{\rho}{A} \frac{\partial L}{\partial s} - \frac{\rho L}{A^2} \frac{\partial A}{\partial s} + \frac{L}{A} \frac{\partial \rho}{\partial s} \quad \dots(25.59)$$

Dividing Eqn. 25.59 throughout by resistance $R = \rho L/A$, we have

$$\frac{1}{R} \frac{dR}{ds} = \frac{1}{L} \frac{\partial L}{\partial s} - \frac{1}{A} \frac{\partial A}{\partial s} + \frac{1}{\rho} \frac{\partial \rho}{\partial s} \quad \dots(25.60)$$

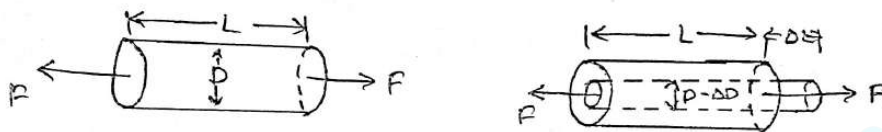
It is evident from Eqn. 25.60, that the per unit change in resistance is due to :

- (i) per unit change in length = $\Delta L/L$, (ii) per unit change in area = $\Delta A/A$, and
- (iii) per unit change in resistivity = $\Delta \rho/\rho$.

Theory :-

If a strip of elastic material is subjected to tension or positively strained, its longitudinal dimension will increase while there will be a reduction in lateral dimension.

So when a gauge is subjected to a positive strain, its length increases, while area of cross section decreases.



Also there will be a change in value of resistivity when conductor is strained.

Consider a strain gauge having dimensions, length = L , area = A , diameter = D before being strained. The material of wire has resistivity ' ρ '

$$\therefore \text{Resistance of unstrained gauge} = R = \frac{\rho L}{A} \quad \text{--- ①}$$

Let a tensile stress ' s ' be applied to the wire which causes length to increase and area to decrease.

$\Delta L \rightarrow$ change in length

$\Delta A \rightarrow$ change in area

$\Delta D \rightarrow$ change in diameter

$\Delta R \rightarrow$ change in resistance.

differentiate R with respect to stress, s

$$\frac{dR}{ds} = \frac{\rho}{A} \frac{\partial L}{\partial s} - \frac{\rho L}{A^2} \frac{\partial A}{\partial s} + \frac{L}{A} \frac{\partial \rho}{\partial s} \quad \text{--- (2)}$$

dividing (2) throughout by resistance $R = \rho L/A$

$$\frac{1}{R} \frac{dR}{ds} = \frac{1}{L} \frac{\partial L}{\partial s} - \frac{1}{A} \frac{\partial A}{\partial s} + \frac{1}{\rho} \frac{\partial \rho}{\partial s} \quad \text{--- (3)}$$

$\therefore$ from (3), the per unit change in resistance is due to,

- i) per unit change in length $= \Delta L/L$
- ii) per unit change in area $= \Delta A/A$
- iii) per unit change in resistivity $= \Delta \rho/\rho$

We know, area, $A = \frac{\pi}{4} D^2$

$$\therefore \frac{\partial A}{\partial s} = 2 \cdot \frac{\pi}{4} \cdot D \cdot \frac{\partial D}{\partial s}$$

$$\text{or } \frac{1}{A} \frac{\partial A}{\partial s} = \frac{(2\pi/4) D}{(\pi/4) D^2} \cdot \frac{\partial D}{\partial s} = \frac{2}{D} \frac{\partial D}{\partial s}$$

$\therefore$ (3) can be written as,

$$\frac{1}{R} \frac{dR}{ds} = \frac{1}{L} \frac{\partial L}{\partial s} - \frac{2}{D} \frac{\partial D}{\partial s} + \frac{1}{\rho} \frac{\partial \rho}{\partial s}$$

Now, poisson's ratio,

$$V = \frac{\text{lateral strain}}{\text{longitudinal strain}} = -\frac{\partial D/D}{\partial L/L}$$

$$\Rightarrow \frac{\partial D}{D} = -V \frac{\partial L}{L}$$

$$\therefore \frac{1}{R} \frac{dR}{ds} = \frac{1}{L} \frac{\partial L}{\partial s} + V \frac{2}{L} \frac{\partial L}{\partial s} + \frac{1}{\rho} \frac{\partial \rho}{\partial s}$$

For small variations, above relation can be

$$\frac{\Delta R}{R} = \frac{\Delta L}{L} + 2\nu \frac{\Delta L}{L} + \frac{\Delta S}{S}$$

The gauge factor is defined as the ratio of change in resistance to per unit change in length.

$$\text{Gauge factor } G_f = \frac{\Delta R/R}{\Delta L/L}$$

$$\Rightarrow \frac{\Delta R}{R} = G_f \frac{\Delta L}{L} = G_f \times \epsilon$$

Where $\epsilon = \frac{\Delta L}{L} \rightarrow \text{strain}$.

The gauge factor can be written as,

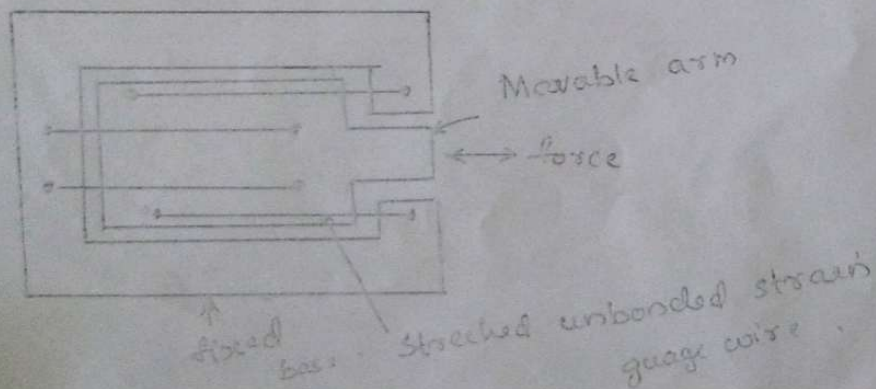
$$= 1 + 2\nu + \frac{\Delta S/S}{\epsilon}$$

$$G_f = \underset{\substack{\downarrow \\ \text{Resistance due} \\ \text{to change in length}}}{1} + \underset{\substack{\downarrow \\ \text{Resistance due to} \\ \text{change in area}}}{2\nu} + \underset{\substack{\text{Resistance change} \\ \text{due to} \\ \text{piezoresistive} \\ \text{effect}}}{\frac{\Delta S/S}{\epsilon}}$$

Types of strain gauges:-

1) unbonded resistance wire strain gauge:-

It consists of a wire stretched between two points in an insulating medium such as air.



The wires are kept under tension so that there is no sag & no free vibration.

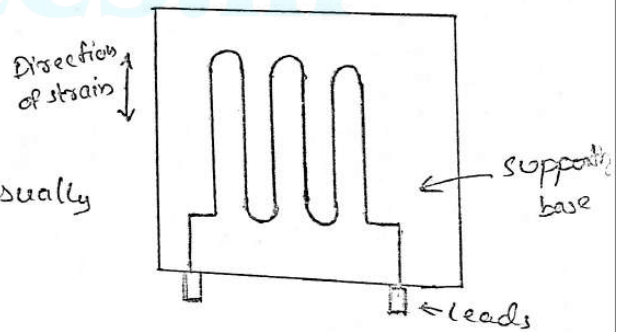
Unbonded strain gauges are usually connected in a bridge circuit & bridge is balanced when no load is applied.

When an external load is applied, resistance of strain gauge changes, causing an unbalance of the bridge circuit resulting in an output voltage.

This voltage is proportional to strain

Bonded wire strain gauges:-

A fine wire element is looped back and forth on a carrier (base) which is usually cemented to the member undergoing stress.

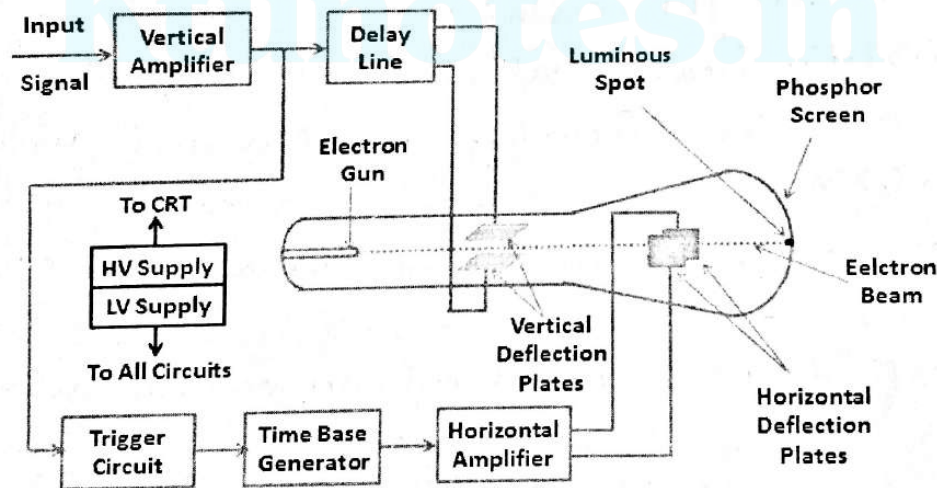


The grid of fine wire is cemented on a carrier which may be a thin sheet of paper, bakelite or teflon. The spreading of wire permits uniform distribution of stress.

Oscilloscope

The Oscilloscope is a basic instrument for study of all types of waveforms. It allows the amplitude of electrical signals, whether they be voltage, current, power etc to be displayed primarily as a function of time. It can be employed to measure quantities such as peak voltage, frequency or period, phase difference, pulse width, delay time, rise time, fall time.

Block diagram.

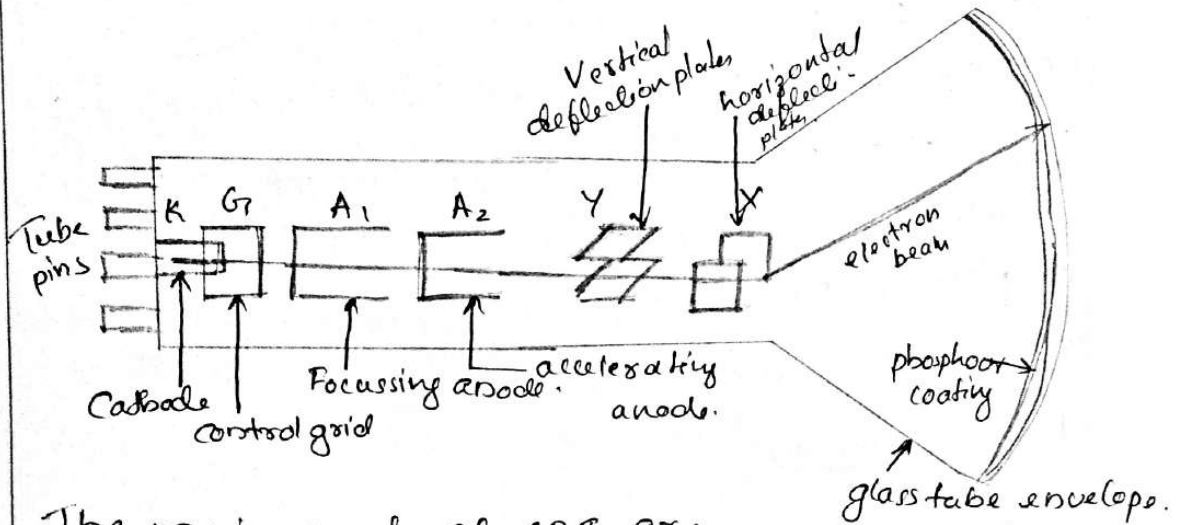


Block Diagram of Cathode Ray Oscilloscope (CRO)

It consist of following elements.

- 1) Cathode Ray tube (CRT) (Basic principle of signal display).
It generates the electron beam, accelerates the beam to a high velocity, deflects the beam

to create a image and contains a phosphor screen where the electron beam eventually become visible. CRT displays the quantity being measured.



The main parts of CRT are

1) Electron gun assembly :- It consist of an indirectly heated cathode, a control grid surrounding the cathode, a focussing anode (A_1) and an accelerating anode (A_2). It is to provide a focused electron beam which is accelerated towards phosphorous screen.

→ Cathode is a nickel cylinder coated with an oxide coating of barium & strontium & emits plenty of electrons when heated. Rate of emission of electrons depends on the magnitude of cathode current, which can be controlled by control grid

→ The control grid is usually a metal cylinder covered at one end but with a small hole in cover, which is provided to allow passage for

→ electrons through it and concentrate the beam along the axis of the tube.

→ Electron beam comes out from control grid through a small hole in it and enters a pre-accelerating anode which is a hollow cylinder in shape and is at a potential of few hundred volts more positive than the cathode so as to accelerate the electron beam.

→ Electron beam is then focused on the screen by focussing anode and accelerating anode. Its function is to focus and accelerate speed of electrons.

→ Electron beam after leaving the electron gun passes through two pairs of deflection plates.

One pair of plates is mounted horizontally and produces an electric field in the vertical plane. This pair produces a vertical deflection and is called vertical deflection plates or Y-plates.

The other pair of plates is mounted vertically and produces a horizontal deflection. & called as horizontal deflection plates or X-plates.

When a sinusoidal voltage is applied to Y plates the beam will be moved up & down according to variation of plate potential.

→ The signal to be displayed on the screen is always applied across the Y plates.

The voltage applied across X-plates is a $\sin$ voltage ie, a voltage which increases linearly with time. It is also called horizontal linebase or sweep voltage



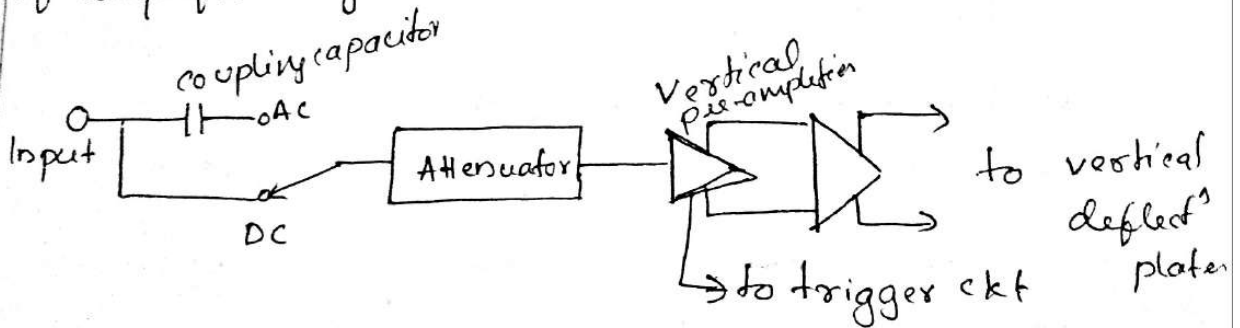
Fluorescent screen:-

The end wall of CRT called screen is coated with phosphor. When struck by high energy e^- , it glows, ie, it absorbs kinetic energy of electrons & convert it into light.

Glass envelope

All the components of a CRT are enclosed in a evacuated glass tube called envelope and it allows the emitted electrons to move freely from one end to other.

2) Vertical deflection system:- The signals to be examined are usually applied to the vertical or y deflection plates through an input attenuator and a number of amplifier stages.



vertical amplifier is required because the signals are not strong enough to produce

measurable deflection on CRT screen. The function of Vertical amplifier is to amplify the input signal. The output stage of amplifier is a pushpull amplifier.

It is necessary to adjust the magnitude of input signal so that a large signal may not cause the display to go out of screen. Therefore input signal is fed to the vertical amplifier through an attenuator.

- The vertical amplifier is the principal factor in determining the sensitivity of oscilloscope.

- Practically all oscilloscopes provide a switchable input coupling capacitor for the measurement of ac signal. When Dc measurements are done, capacitor may be removed.

5) Horizontal deflection system:-

The horizontal deflection plates (X plates) are fed by a sweep voltage that provides a time base. The horizontal plates are supplied through an amplifier whose function is to amplify the signal to X plates.

For measurement of voltages and currents, the input to the horizontal amplifier is a sawtooth wave. For frequency and phase measurements, the input is an external signal.

The horizontal amplifier should be capable of amplifying both these inputs without distortion.

- when external signals are to be applied to the horizontal deflection system, they can be fed through the horizontal amplifier via the sweep selector switch in the external position.

- when the sweep selector switch is in the internal position, the horizontal amplifier receives an input from the sawtooth sweep generator.

Basic Sweep generator:

If the horizontal deflection plates, X-plates are not energised, the electron beam would draw a vertical line on the screen.

→ The function of time base generator is to drive the beam at a steady speed across the screen and when it reaches the right hand side of the screen, the beam is made to flyback to the starting position.

→ The time base generator produces a sawtooth wave of the same frequency as the input signals to the Y-plates

→ The horizontal deflection of the beam is proportional to the instantaneous voltage of the sawtooth wave

→ The line base generator gives an output sweep signal of adjustable frequency

Types of sweeps are

i) free running sweep → it will be a repetitive sawtooth waveform i.e., a new sweep starts immediately after the previous sweep

ii) Triggered sweep: Sometimes the waveform to be observed may not be periodic, so sweep is initiated by waveform under examination. In such cases we use triggered sweep.

iii) Driven sweep: Here, the sweep is free running but triggered by signal under test

iv) Non sawtooth sweep: For comparing two frequencies or finding phase shift, non sawtooth sweep is utilised.

Delay line:

The purpose of a delay line is to delay the vertical signal enough to keep it from reaching the CRT deflection plates before the horizontal sweep circuits are running.

In an oscilloscope, all electronic circuitry causes a certain amount of delay in the transmission of signal voltages to the deflection plates.

or may be of several selectable magnitudes. Provision of this facility enables the oscilloscope amplifiers and time base to be calibrated and checked for accuracy.

11.20.4. Cameras

The use of photography is the best method of permanent records of oscilloscope traces for analysis.

- The special cameras used for keeping the records are of the following two types :
 - (i) 35 mm. film;
 - (ii) Polaroid film.
- 35 mm unit is usually prefocused and bolted over the tube face. Some cameras have a fixed lens aperture, the exposure is simply controlled by hand operation of flapper shutter.
- Polaroid film photographic technique approaches instant photography. A permanent record is obtained within 10–20 s. However, this process is comparatively expensive.

11.21 CRO MEASUREMENTS *(Application of CRO)*

CRO can be used to measure various characteristics of an input signal and the properties of the output signal. The various parameters which can be measured by using CRO are :

- | | |
|--------------------------|------------------------|
| (i) Voltage; | (ii) Current; |
| (iii) Period; | (iv) Frequency; |
| (v) Phase; | (vi) Amplitude; |
| (vii) Peak-to-peak value | (viii) Duty cycle etc. |

11.21.1. Voltage measurement

The procedure for voltage measurement is as follows :

- The voltage to be measured is applied to Y-deflection plates through the vertical amplifier. Generally, a coaxial cable is used for connection.
- The X-deflection plates are excited by the time-base generator.
- To aid the measurement, a set of squares is marked on the transparent screen cover. Using these markings and the gain of the vertical amplifier, the peak-to-peak (p-p) value (and hence r.m.s. value) of the voltage wave can be measured.
- After noting down the selection in *volts/division* from the front panel (selected for measurement) the peak-to-peak value, amplitude and r.m.s value of sinusoidal signal can be obtained as given below :

(i) Peak-to-peak value, $V_{p-p} = \left(\frac{\text{Volts}}{\text{div.}} \right) \times (\text{number of divisions})$

(The volts/div. is nothing but deflection sensitivity of CRO)

(ii) Amplitude, $V_{max} = \frac{V_{p-p}}{2}$

(iii) R.M.S. value, $V_{r.m.s.} = \frac{V_{p-p}}{2\sqrt{2}}$ (for sinusoidal signals only)

- It must be remembered that an oscilloscope is very complex device and that the accuracy of measurements is not better than $\pm 3\%$. It is basically a *display device* so that we can observe the waveform in detail and find out the extent of harmonics in the wave.

Example 11.7. For the waveform shown in Fig. 11.29, if vertical attenuation is 3 mV/div. determine :

- Peak-to-peak value,
- Amplitude, and
- R.M.S. value of the signal.

Solution. Vertical attenuation = 3 mV/div.

From the figure, it is evident that the screen is divided such that one part is subdivided into 5 units.

$$\therefore 1 \text{ subdivision} = \frac{1}{5} = 0.2 \text{ units (see Fig. 11.29)}$$

- (i) Peak-to-peak value- V_{p-p} :

It can be observed that positive peak of the signal corresponds to two full divisions and three subdivisions. Hence positive peak is $2 + 3 \times 0.2 = 2.6$ units. While the negative peak also corresponds to 2.6 units.

$$\therefore V_{p-p} = 2.6 + 2.6 = 5.2 \text{ divisions}$$

$$\begin{aligned} \text{Also, } V_{p-p} &= \frac{\text{Volts}}{\text{division}} \times \text{No. of divisions} \\ &= 3 \times 5.2 \\ &= 15.6 \text{ mV (Ans.)} \end{aligned}$$

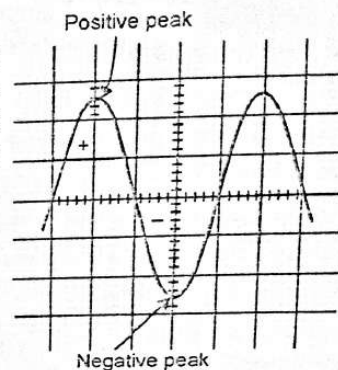


Fig. 11.29.

(ii) Amplitude, V_{max} :

$$V_{max} = \frac{V_{p-p}}{2} = \frac{15.6}{2} = 7.8 \text{ mV (Ans.)}$$

(iii) R.m.s. value, $V_{r.m.s.}$:

$$V_{r.m.s.} = \frac{V_{max}}{\sqrt{2}} = \frac{7.8}{\sqrt{2}} = 5.515 \text{ mV (Ans.)}$$

11.21.2. Current measurement

A CRO has a very high input impedance and cannot be used for direct measurement of current. However, the current can be measured in terms of voltage drop across a standard resistance as shown in Fig. 11.30; i.e., $I = \frac{V \text{ (measured on CRO)}}{R}$.

- For measurement of high frequency currents, the shunt should be non-inductive.

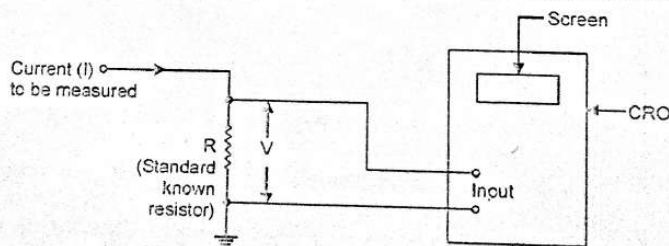


Fig. 11.30. Current measurement.

loads upto 25 kN with an accuracy of 0.1%.

7.14.3. Strain-Gauge Load Cells :-

These cells convert weight or force into electrical outputs which are provided by strain gauges; these outputs can be connected to various measuring instruments for indicating, recording and controlling the weight or force.

Usually the strain gauges are directly applied to the force-developing device and the device is calibrated against strain-gauge output.

- These are excellent force-measuring devices, particularly for transient and steady forces.
- These are used in conjunction with CRO (for display purposes) for measurement of rapidly changing loads.

Construction and working of the load cell :

Fig. 7.50 shows a simple strain gauge load cell. It consists of a steel cylinder, on which are mounted four identical strain gauges. The gauges R_{g1} and R_{g4} are along the direction of applied load and the gauges R_{g2} and R_{g3} are attached circumferentially to the cylinder. All the four gauges are connected electrically to the four limbs of a Wheatstone bridge circuit.

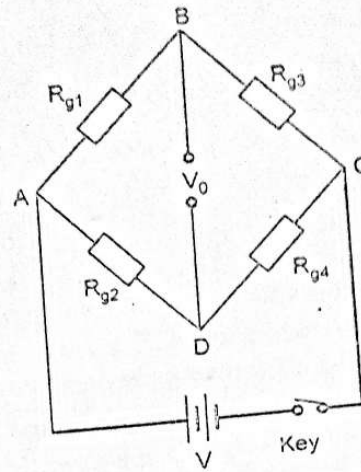
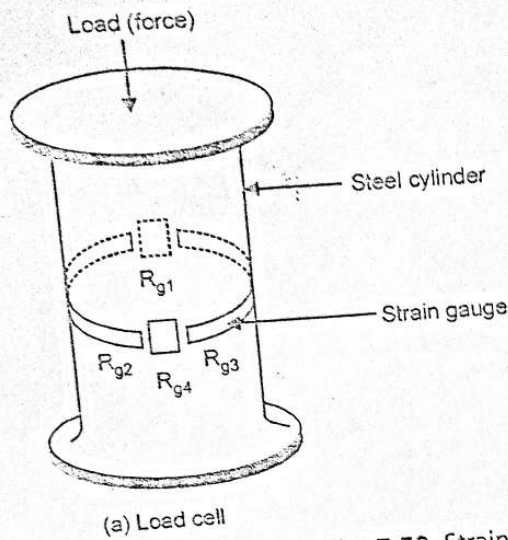


Fig. 7.50. Strain gauge load cell.

When there is no load on the cell, all the four gauges have the same resistance (i.e., $R_{g1} = R_{g2} = R_{g3} = R_{g4}$). Obviously the terminals B and D are at the same potential, the bridge is balanced and the output voltage is zero.

...(7.66)

$$V_{AB} = V_{AD} = \frac{V}{2}$$

...(7.67)

$$V_{AB} - V_{AD} = V_0 = 0$$

On the application of a compressive load to the unit, the vertical gauges (R_{g1} and R_{g3}) undergo compression (i.e., negative strain) and, therefore, there is decrease in resistance. The circumferential gauges R_{g2} and R_{g4} , simultaneously, undergo tension (i.e., positive strain) and, therefore, there is increase in resistance. The two strains are not equal; these are related to each other by a factor, μ , the Poisson's ratio. Thus when strained, the resistances of various gauges are:

$$R_{g1} = R_{g3} = R - dR$$

...(compression)

$$R_{g2} = R_{g4} = R + dR$$

...(tension)

$$\begin{aligned} \text{Potential at terminal B, } V_{AB} &= \frac{R_{g1}}{R_{g1} + R_{g3}} V = \frac{R - dR}{(R - dR) + (R + \mu dR)} \times V \\ &= \frac{R - dR}{2R - dR(1 - \mu)} \times V \end{aligned}$$

...(i)

$$\begin{aligned} \text{Potential at terminal D, } V_{AD} &= \frac{R_{g2}}{R_{g2} + R_{g4}} V = \frac{R + \mu dR}{(R + \mu dR) + (R - dR)} \times V \\ &= \frac{R + \mu dR}{2R - dR(1 - \mu)} \times V \end{aligned}$$

...(ii)

The changed output voltage,

$$V_0 + dV_0 = \frac{R - dR}{2R - dR(1 - \mu)} \times V - \frac{R + \mu dR}{2R - dR(1 - \mu)} \times V$$

...[Using (i) and (ii)]

$$= \frac{dR(1+\mu)}{2R} = 2(1+\mu) \left(\frac{dR}{R} \cdot \frac{V}{4} \right)$$

Since the output voltage $v_0 = 0$ under unloaded conditions, therefore, change in voltage due to applied load becomes :

$$dv_0 = 2(1+\mu) \left(\frac{dR}{R} \cdot \frac{V}{4} \right)$$

Obviously, this voltage is a measure of the applied load.

The use of four identical strain gauges in each arm of the bridge provides full temperature compensation and also increases the sensitivity of the bridge $2(1+\mu)$ times.

Uses : The strain gauge load cells find extensive use in the following :

- (i) Road vehicle weighing devices.
- (ii) Draw bar and tool-force dynamometers.
- (iii) Crane load monitoring etc.

Example 7.22. The following data relate to strain gauge load cell arranged with four strain gauges as shown in Fig. 7.50.

Diameter of the steel cylinder = 60 mm; Nominal resistance of each gauge = 120 Ω ; Factor = 2.0; Supply voltage (V) = 6V; Modulus of elasticity for steel = 200 GN/m²; Poisson's ratio = 0.3.

Calculate the sensitivity of the load cell.

Solution. Given : $d = 60$ mm = 0.06 m; R_g (each gauge) = 120 Ω ; $G_f = 2.0$; $E = 200$ GN/m²; $\mu = 0.3$.

Sensitivity of the load cell :

Consider a load of 1 kN applied to the load cell.

$$\text{Stress } (\sigma) = \frac{\text{Load}}{\text{Cross-sectional area}} = \frac{1 \times 10^3}{\frac{\pi}{4} \times (0.06)^2} = 0.3537 \times 10^6 \text{ N/m}^2$$

$$\text{Strain, } e = \frac{\text{Stress } (\sigma)}{\text{Modulus of elasticity } (E)} = \frac{0.3537 \times 10^6}{200 \times 10^9} = 1.7685 \times 10^{-6}$$

Fraction change in resistance,

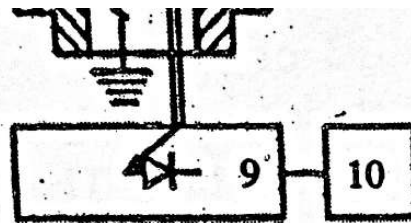
$$\frac{dR}{R} = G_f \times e = 2.0 \times 1.7685 \times 10^{-6} = 3.537 \times 10^{-6}$$

$$\text{Output voltage, } dv_0 = 2(1+\mu) \left(\frac{dR}{R} \cdot \frac{V}{4} \right)$$

...[Equation]

$$= 2(1+0.3) \left(3.537 \times 10^{-6} \times \frac{6}{4} \right) = 13.79 \times 10^{-6} \text{ V} = 13.79 \text{ } \mu\text{V}$$

Hence, the sensitivity of the load cell = 13.79 $\mu\text{V/kN}$ (Ans.)



- resistance
- 9) Digital-analog converter
 - 10) Indicating or recording unit

Figure 3.8: Current Transformer with Electro-Optical Signal Converter for EHV Systems

Ques 6) What do you mean by PMU? Also list its applications in power system.

Or

Briefly discuss the term "Phasor Measurement Unit".

Or

Explain the concept of phasor measurement.

Ans: Phasor Measurement Unit (PMU)

PMUs technology provides phasor information (both magnitude and phase angle) in real time. The advantage of referring phase angle to a global reference time is helpful in capturing the wide area

snapshot of the power system. Effective utilization of this technology is very useful in mitigating blackouts and learning the real time behavior of the power system.

With the advancement in technology, the microprocessor based instrumentation such as protection Relays and Disturbance Fault Recorders (DFRs) incorporate the PMU module along with other existing functionalities as an extended feature.

A pure sinusoidal waveform can be represented by a unique complex number known as a phasor. Consider a sinusoidal signal

$$x(t) = X_m \cos(\omega t + \phi) \quad \dots(1)$$

The phasor representation of this sinusoidal is given by

$$x(t) = \frac{X_m}{\sqrt{2}} e^{j\phi} = \frac{X_m}{\sqrt{2}} (\cos\phi + j\sin\phi) \quad \dots(2)$$

Note that the signal frequency ω is not explicitly stated in the phasor representation. The magnitude of the phasor is the rms value of the sinusoid $\frac{X_m}{\sqrt{2}}$ and its phase angle is ϕ , the phase angle of the signal in (1). The sinusoidal signal and its phasor representation given by (1) and (2) are illustrated in figure 3.9(b).

Note that positive phase angles are measured in a counterclockwise direction from the real axis. Since the frequency of the sinusoidal is implicit in the phasor definition, it is clear that all phasors which are included in a single phasor diagram must have the same frequency.

Phasor representation of the sinusoidal implies that the signal remains stationary at all times, leading to a constant phasor representation. These concepts must be modified when practical phasor measurements are to be carried out when the input signals are not constant, and their frequency may be a variable.

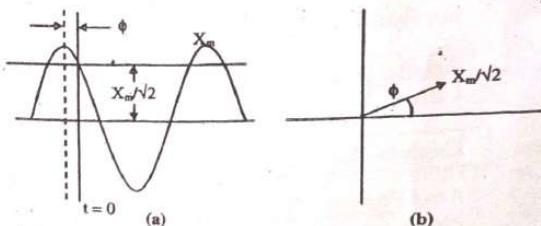


Figure 3.9: Phasor Representation of a Sinusoidal Signal.
a) Sinusoidal Signal b) Phasor Representation

Concept of Phasor Measurement

Although a constant phasor implies a stationary sinusoidal waveform, in practice it is necessary to deal with phasor measurements which consider the input signal over a finite data window. In many PMUs the data window in use is one period of the fundamental frequency of the input signal.

If the power system frequency is not equal to its nominal value (it is seldom), the PMU uses a frequency-tracking step and thus estimates the period of the fundamental frequency component before the phasor is estimated. It is clear that the input signal may have harmonic or non-harmonic components. The task of the PMU is to separate the fundamental frequency component and find its phasor representation.

The most common technique for determining the phasor representation of an input signal is to use data samples taken from the waveform, and apply the Discrete Fourier Transform (DFT) to compute the phasor. Since sampled data are used to represent the input signal, it is essential that antialiasing filters be applied to the signal before data samples are taken.

The antialiasing filters are analog devices which limit the bandwidth of the pass band to less than half the data sampling frequency (Nyquist criterion), if x_k ($k = 1, 2, 3, \dots, N-1$) are the N samples of the input signal taken over one period, then the phasor representation is given by

$$X = \frac{\sqrt{2}}{N} \sum_{k=0}^{N-1} x_k e^{-jk\frac{2\pi}{N}} \quad \dots(3)$$

The multiplier in front of the summation sign may need some explanation. Note that for real input signals, the components of the signal at a frequency ω appear in the DFT at $\pm\omega$ and are complex conjugates of each other. They can be combined, giving a factor of 2 in front of the summation sign in (3). The peak value of the frequency thus obtained is then converted to rms value by dividing by $\sqrt{2}$.

The DFT calculation eliminates the harmonics of the input signal. However, the non-harmonic signals and any other random noise present in the input signal leads to an error in estimation of the phasor.

Synchrophasor is a term used to describe a phasor which has been estimated at an instant known as the time tag of the synchrophasor. In order to obtain simultaneous measurement of phasors across a wide area of the power system, it is necessary to synchronize these time tags, so that all phasor measurements belonging to the same time tag are truly simultaneous.

the most popular types as of this writing are the light-emitting diode (LED) and liquid crystal diode (LCD) displays.



Fig. 10.4. Seven-segment decimal readout.

- Another type of display is 5×7 dot matrix, shown in Fig. 10.5 :

- It consists of 35 light sources (usually LEDs) arranged in *five vertical columns*, each containing *seven light sources*.
- This 5×7 matrix of light sources can be illuminated to form all numerical and alphabetic characters.

Applications of dot matrix method :

- Athletic score board;
- Bank time-temperature billboards;
- Cathode ray tube (CRT), computer TV displays etc.

Fig. 10.5. 5×7 dot matrix.

10.3.5. Frequency Counters

Refer to Fig. 10.6 :

- Fig. 10.6(a) shows the basic block diagram of a "frequency counter". It consists of the following sections :

- Decode counter assembly (DCA);
- Main gate;
- Trigger;
- Main gate flip-flop;
- Display clock.

- The frequency counter measures "events per unit of time (EPUT)" (i.e., cycles per second), so the DCA must be tuned on only for a given period of time (e.g., 0.1, 1, or 10 s).

- The main gate, main gate flip-flop, and time base sections are used to allow input pulses into the DCA for the designed period of time.

- The time base section consists of a crystal oscillator that produces pulses at precise rate such as 100 kHz, 1 MHz, 4 MHz, 10 MHz, and so on. A chain of *decade dividers* is used to reduce the crystal oscillator frequency to a *lower frequency*. The time base output frequency will be 10 Hz for 0.1 s, 1 Hz for 1 s and 0.1 Hz for 10 s measuring periods.

- Fig. 10.6(b) shows the timing diagram for one complete interval of EPUT counter.

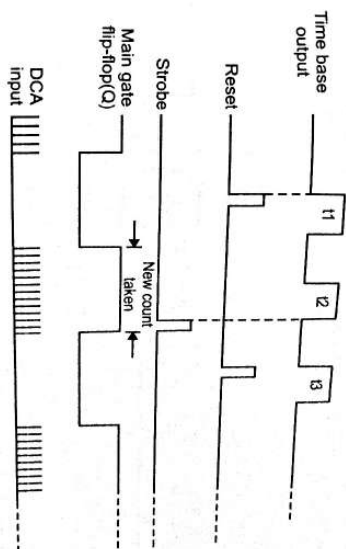
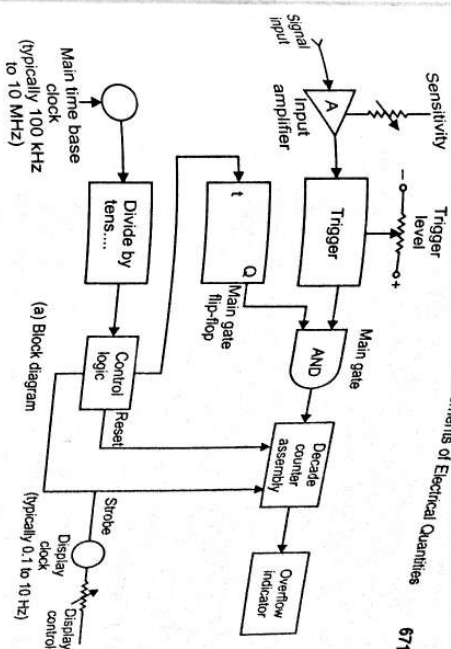


Fig. 10.6. Frequency counter.

- Pulses t_1 , t_2 and t_3 are output from the time base section. When pulse t_1 goes high, the control logic section generates a short pulse to reset the DCA to zero. When t_1 goes low again, the Q output of the J-K main gate flip-flop will go high. The main (AND) gate has one input tied to the Q output of the flip-flop and the other input is tied to the signal being counted. As a result, the main gate passes input pulses to the DCA *only* when the Q terminal of the flip-flop is high.

- Until the negative transition of t_2 occurs, the flip-flop remains set. At that time the Q output of the flip-flop drops low, turning off the flow of pulses into the DCA and causing the control logic section to generate a strobe pulse. This pulse tells the DCU latches to transfer data from the counter to the decoders. The display, then, shows *only completed count cycles and will hold the previous count until the end of the next interval*.

- The frequency of the input signal is the number of counts accommodated on the DCA divided by the time base period in seconds.

i.e.,

$$f_{(Hz)} = \frac{\text{Counts on DCA}}{\text{Time (s)}}$$

Note If the time base is 1s, the DCA reading will be frequency in Hz, and if the time base is a decade multiple or submultiple of 1 Hz, then the frequency in Hz can be determined by adjusting the decimal point.

Example 10.1. A digital frequency counter which has a 3-digit display, gated period in milliseconds, is selected to measure an unknown frequency. The reading is 045. What is the frequency of the system?

Solution. Given : Count, $N = 045$

Gated period, $t = 10 \text{ ms}$

Now, frequency of the system,

$$f = \frac{N}{t} = \frac{045}{10 \times 10^{-3}} = 4500 \text{ Hz or } 4.5 \text{ kHz (Ans.)}$$

999 upto 1999, to give overlap between ranges for convenience, a process called "over-ranging". This type of display is called a $3\frac{1}{2}$ digit display, shown in Fig. 10.8.

10.3.8. Analog-to-Digital (A/D) and Digital-to-Analog Conversion

Refer to Article 9.4

10.4 DIGITAL VOLTMETERS (DVMs)

10.4.1. Introduction

A digital voltmeter (DVM) displays measurements of D.C. or A.C. voltage as discrete numerals instead of pointer deflections on a continuous scale as in analog instruments.

- It is a versatile and accurate instrument that is employed in many laboratory measurement applications.
- Owing to the development and perfection of IC modules, the size, power requirements and cost of the digital voltmeter have been drastically reduced and, therefore, DVMs can actively compete with conventional analog instruments, both in price and portability.
- The various varieties of DVM differ in the following ways :
 - (i) Number of digits;
 - (ii) Number of measurements;
 - (iii) Accuracy;
 - (iv) Speed of reading;
 - (v) Digital output of several types.

10.4.2. Advantages of DVMs

Following are the advantages of DVM over other voltmeters :

1. Higher accuracy and resolution.
2. Greater speed.
3. No parallax.
4. Reduced human error.
5. Compatibility with other digital equipment for further processing and recording.

10.4.3. Characteristic Features of DVMs

The characteristic features of digital voltmeters (DVMs) are :

1. **Input range** : From ± 1.000 V to ± 1000 V with automatic range selection and overload indication.
2. **Absolute accuracy** : As high as $\pm 0.005\%$ of the reading.
3. **Resolution** : 1 part in million ($1 \mu\text{V}$ reading can be read or measured on 1 V range)
4. **Stability** : Short-term, 0.002% of the reading for a 24-hr period; long term, 0.008% of the reading for a 6-month period.
5. **Input resistance** : Typically 10 M Ω ;
Input capacitance : 40 pF
6. **Calibration** : Internally from stabilised reference sources, independent of measuring circuit.
7. **Output signals** : In BCD (Binary-coded-decimal) form, for print output and further digital processing.
- **Optional features** may include additional circuitry to measure current, ohms and voltage ratio.

- (i) Op-amp, used as an integrator;
- (ii) A level comparator;
- (iii) A basic block, for generating timing pulses;
- (iv) A set of decimal counters;
- (v) A block of logic circuitry.

(v) The unknown voltage V_x is applied through switch S to the integrator for a known period of time T as shown in Fig. 10.11. This period is determined by counting the clock frequency in decimal counters. During time period T , C is charged at the rate proportional to V_x .

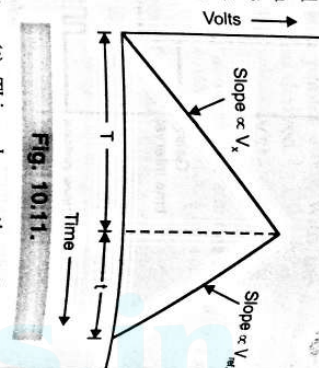


Fig. 10.11.

At the end of the interval T , S is shifted to the reference voltage V_{ref} of opposite polarity. The capacitor charge begins to decrease with time and results in downward linear ramp voltage. During the second period a known voltage (i.e., V_{ref}) is observed for an unknown time t is determined by counting timing pulses from the clock until the voltage across the capacitor reaches its basic reference value (reference may be ground or any other basic reference level).

Then, from similar triangles of Fig. 10.11 we have :

$$\frac{V_x}{T} = \frac{V_{ref}}{t} \quad \dots(10.5)$$

$$V_x = \frac{V_{ref}}{t} \times T$$

The count after t which is proportional to the input voltage V_x is displayed as the measured voltage.

• This instrument can be used to measure currents, resistances and A.C. voltages by using appropriate signal conditioners.

Advantages. The averaging characteristics and cancellation of errors that usually limit the performance of ramp-type DVM are the main advantages of such DVMs:

- The integration characteristics provide the average value of the input signal during the period of first integration. Consequently, disturbance, such as spurious noise pulses, are minimised.
- Long-term drifts in the time constant, as may result from temperature variations or aging, do not affect conversion accuracy.
- The long-term alterations in clock frequency have no effect.

10.4.8. Integrating Type DVM (Voltage-to-frequency conversion)

This voltmeter indicates the true average value of the unknown voltage over a fixed measuring period.

It employs an "integration technique" which uses a voltage-to-frequency (V/F) conversion. The V/F converter functions as a feedback control system which governs the rate of pulse generation in proportion to the magnitude of the input voltage.

In V/F conversion technique, a train of pulses, whose frequency depends upon the voltage being measured, is generated. Then the number of pulses appearing in a definite interval of time is measured.

The basic requirement of this technique is "operational integrator", the circuit diagram and symbolic representation of which is shown in [Fig. 10.12(a, b)]. It is a modification of operational amplifier. The operational integrator consists of an amplifier (having high gain) with a capacitor C connected between output and input. If V_i is the input voltage, then V_o of the operational integrator is given by :

$$V_o = -\frac{1}{RC} \int V_i \cdot dt \quad \dots(10.6)$$

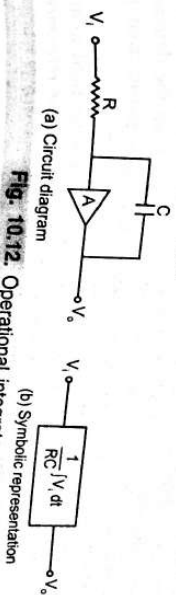


Fig. 10.12. Operational integrator.

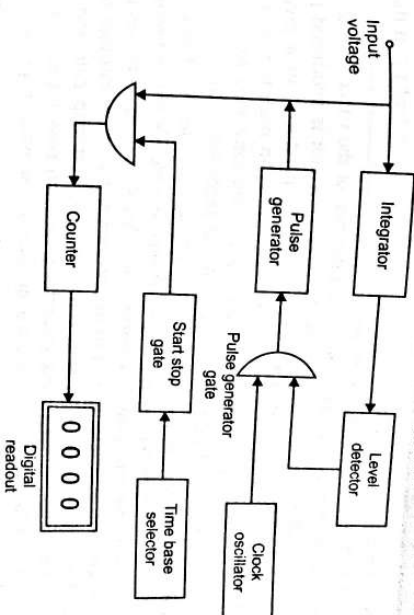


Fig. 10.13. Block diagram of an integrating type DVM (Voltage-to-frequency conversion).

If the input voltage is constant, we have :

$$V_o = -\frac{V_i}{RC} \cdot t \quad \dots(10.7)$$

Thus, if a constant input voltage V_i is applied, an output voltage V_o is produced which rises at a uniform rate and has a polarity opposite to that of input voltage. That is, for a constant input voltage the integrator produces a ramp output voltage of opposite polarity such that the greater the input voltage, the sharper is the rate of rise, or step of the output voltage.

Fig. 10.13. shows the block diagram of this voltmeter :

- The unknown voltage, V_i is applied to the input of the integrator and the output voltage V_o starts to rise. The slope of V_o is determined by the value of input voltage V_i . This voltage is fed to a level detector, and, when V_o reaches a certain reference level, the detector sends a pulse to pulse generator gate. The level detector is a device similar to a voltage comparator.
- The output voltage from integrator, V_o is compared with the fixed voltage of an internal reference source and when V_o reaches that level, the level detector

produces an output pulse. This output pulse opens the pulse level gate, permitting pulse from a fixed frequency clock oscillator to pass through the pulse generator. This generator is similar to schmitt trigger that produces an output pulse of fixed amplitude and width for every pulse it receives.

- The output pulse whose polarity is opposite to that of V_i and has greater amplitude is fed back to the input of the integrator. Hence, the net output to the integrator is now of reversed polarity (opposite to that of V_i). As a result of this reversed output, the output V_o drops back to its original level. Since V_o is now below the reference level of the level detector, there is no output from the detector to the pulse generator gate and the gate gets closed. Thus, no more pulses from the clock oscillator can pass through to trigger the pulse generator.
- When the output voltage pulse generated has passed, V_i is restored to its original value and V_o starts rising again. When it reaches the level of reference voltage again, the pulse generator gate is opened. The pulse generator is triggered by a pulse from the clock generator and the entire cycle is repeated again. Thus, the waveform of V_o is sawtooth wave whose rise time is dependent upon the value of the output pulse from the pulse generator.
- Thus, the frequency of the sawtooth V_n is a function of the value V_i , the voltage being measured. Since one pulse from the pulse generator is produced for each cycle of the sawtooth wave, the number of pulses produced in a given time interval and hence the frequency of sawtooth wave is an indicator of the value of voltage being measured. *The frequency of sawtooth wave may be measured by counting the number of pulses in a given interval of time.*
- Pulses from clock oscillator are applied to a 'time base selector'. The first pulse passes through start-stop gate, producing an output which is applied to the main gate, thus opening the gate. As a result of this, the same output pulses from the pulse generator (that are applied to the integrator) also pass through the main gate. The next pulse from the time base closes the start-stop gate and also the main gate. Thus, no more pulse generator pulses can pass through. Hence the counters and their associated readouts indicate the number of pulses that have passed during a known interval of time. This count is an indication of the voltage being measured. *In order to make the counter read directly in terms of voltage the amplitude and width of pulse generator can be suitably adjusted.*
- The drawback of this DVM is that it needs excellent characteristics in the linearity of the ramp. The A.C. noise and supply noise are averaged out.

10.4.9. Successive-approximation DVM

Fig. 10.14, shows the block diagram of a successive-approximation DVM, in which the electronic implementation of the successive-approximation technique is illustrated :

- A "digital-to-analog (D/A) converter" is used to provide the estimates. The "equal to or greater than" or "less than" decision is made by the "comparator". The D/A converter provides the estimate and is compared to the input signal.
- A special shift register called a "successive-approximation register (SAR)" is used to control the D/A converter and consequently the estimates. At the beginning of the conversion all the outputs from the SAR are at logic zero. If the estimate is greater than the input, the comparator output is high and the first SAR output reverses state and the second output changes to a logic "one". If the comparator output is low, indicating that the estimate is lower than the input signal the first-

... junction' is useful for selecting the components within a specified range.

0.6 DIGITAL MULTIMETER (DMM)

Digital multimeter offers *increased versatility due to its additional capability to measure A.C. voltage and current, D.C. current, and resistance.*

Fig. 10.16, shows the block diagram of a digital multimeter (DMM) :

- In the "A.C. voltage mode", the applied input is fed through a calibrated, compensated attenuator, to a precision full-wave rectifier circuit followed by a ripple reduction filter. The resulting D.C. is fed to ADC and the subsequent display system.
- For *current measurements*, the drop across an internal calibrated shunt is measured, directly by the ADC in the "D.C. current mode", and after A.C. to D.C. conversion in the "A.C. current mode". This drop is often in the range of 200 mV (corresponding to full scale).
- Due to the lack of precision in the A.C.-D.C. conversions, the accuracy in the A.C. range is generally of the order of 0.2 to 0.5%. In addition, the measurement range is often limited to about 50 Hz at the lower frequency

end due to the ripple in the rectified signal becoming a non-negligible percentage of the display and hence results in fluctuation of the displayed number. At the higher frequency end, deterioration of the performance of the AC/DC converter limits the accuracy.

— The A.C. measurement, is often average reading, r.m.s. calibrated.

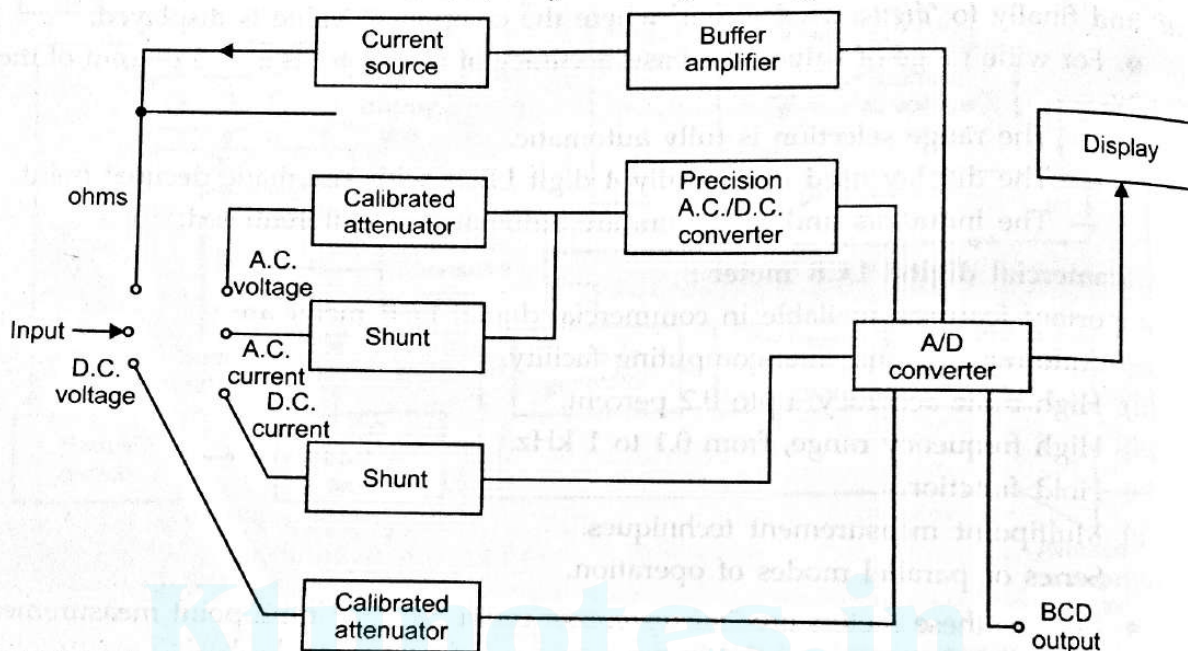


Fig. 10.16. Block diagram of a digital multimeter.

- In the "resistance range" the digital multimeter operates by measuring the voltage across the externally connected resistance, resulting from a current forced through it from a calibrated internal current source.

— The accuracy of resistance measurement is of the order of 0.1 to 0.5% depending on the accuracy and stability of the internal current sources. The accuracy may be proper in the highest range which is often about 10 to 20 MΩ. In the lowest range, the full scale may be $\approx 200 \Omega$ with a resolution of about 0.01Ω for a $4\frac{1}{2}$ digit digital multimeter. In this range of resistance measurement, the effect of the lead resistance will have to be carefully considered.

Comparison between "Analog" and "Digital" Multimeters

S. No.	Aspects	Analog multimeters	Digital multimeters
1.	Power supply	Not required	Required
2.	Visual indication of changes in the reading	Better	Not that much better
3.	Effect of electric noise	Less	More
4.	Isolation problems	Less	More
5.	Accuracy	Less	High